

MCARTHUR RIVER MINE, Australia

WMO: 942390

Lat: 16.442S

Lon: 136.076E

Elev: 135

StdP: 14.62

Time Zone: 9.50 (AUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	47.8	50.8	13.9	11.2	81.7	18.7	14.2	80.0	16.0	72.5	14.5	75.0	3.3	260	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.7	104.1	70.7	102.4	70.8	100.6	71.7	81.3	86.7	80.6	86.9	79.9	86.6	8.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	158.6	82.4	79.3	152.7	82.3	78.4	148.4	82.2	45.2	86.4	44.4	86.9	43.7	86.6	94.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	14.0	12.7	43.0	107.7	2.2	2.2	41.4	109.3	40.1	110.6	38.9	111.9	37.3	113.5		
(5)			37.9	83.4	2.3	3.6	36.3	86.0	34.9	88.2	33.6	90.2	32.0	92.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	86.4	86.0	85.2	82.2	76.7	71.3	71.0	73.2	80.0	85.3	88.0	88.3
	DBStd	7.50	3.68	3.39	3.06	3.35	5.18	5.39	5.10	4.91	4.13	3.56	3.20	3.80
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	22	0	0	0	0	1	9	10	2	0	0	0	0
	CDD50	11354	1130	1008	1092	965	827	640	651	719	901	1095	1139	1187
	CDD65	5902	665	588	627	515	363	200	196	256	451	630	689	722
	CDH74	77052	8649	7524	7770	6277	4166	2279	2502	3483	5868	8743	9716	10075
	CDH80	41424	4566	3936	3977	3277	1917	839	966	1629	3300	5236	5888	5893
Wind	WSAvg	5.8	5.6	5.3	5.1	5.2	5.5	5.7	5.5	5.9	6.3	6.7	6.4	6.0
Precipitation	PrecAvg	28.7	7.0	7.3	5.6	1.1	0.1	0.1	0.1	0.0	0.2	0.6	2.0	5.1
	PrecMax	46.3	17.2	26.0	16.4	7.2	0.7	0.7	2.2	0.0	3.5	3.7	5.8	17.1
	PrecMin	16.5	0.4	0.2	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.9
	PrecStd	8.6	4.6	5.3	4.0	1.5	0.2	0.2	0.4	0.0	0.7	0.9	1.5	3.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.2	103.3	103.3	98.9	95.5	91.6	92.0	95.6	100.1	104.3	106.0	107.7
		MCWB	73.3	70.7	70.9	70.3	69.4	67.8	66.8	66.6	67.0	67.6	70.4	68.6
	2%	DB	101.6	100.3	99.5	96.7	93.3	89.8	89.8	92.9	98.0	101.9	103.7	104.1
		MCWB	74.1	73.8	72.7	70.3	68.5	66.7	65.0	65.5	67.1	67.9	70.2	71.7
	5%	DB	98.9	98.1	97.0	95.0	91.5	87.8	87.9	90.9	96.1	100.0	101.6	101.6
		MCWB	75.3	74.3	73.9	70.5	68.2	65.3	63.7	64.4	66.8	68.9	70.5	72.7
	10%	DB	96.0	95.6	94.8	93.3	89.5	85.3	85.9	88.7	94.2	98.1	99.5	98.9
		MCWB	76.5	75.5	74.2	70.3	67.3	63.6	62.3	62.3	66.0	69.3	71.4	74.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.2	81.6	81.4	79.2	77.5	71.7	72.3	72.9	77.6	79.1	82.2	82.4
		MCDB	87.0	86.9	88.4	83.9	84.6	79.4	85.1	84.7	85.3	84.2	84.4	84.9
	2%	WB	81.0	80.7	80.2	77.8	75.6	70.2	70.3	71.1	74.9	77.1	79.7	80.9
		MCDB	87.3	87.2	87.3	83.8	83.2	80.7	81.5	82.0	83.5	86.0	85.4	85.3
	5%	WB	80.4	80.0	79.6	76.7	74.1	68.7	68.5	69.3	73.3	75.9	78.5	79.9
		MCDB	87.3	86.8	86.5	83.8	82.7	81.0	78.9	81.4	84.1	85.5	85.6	85.7
	10%	WB	79.8	79.3	78.9	75.8	72.6	67.2	66.6	67.3	71.9	75.0	77.4	79.0
		MCDB	87.0	86.3	85.6	83.7	82.1	80.4	78.7	80.4	83.4	85.2	86.0	86.1
Mean Daily Temperature Range	5% DB	MDBR	16.6	17.3	18.3	22.7	25.4	27.3	29.5	31.1	30.0	26.9	23.7	19.7
		MCDBR	22.2	22.5	22.8	24.7	26.0	27.7	30.2	31.9	31.1	28.8	27.5	24.9
		MCWBR	5.9	5.9	5.9	7.1	8.0	9.6	10.1	9.9	8.2	8.2	7.2	6.8
	5% WB	MCDBR	16.5	16.0	16.3	18.7	21.5	24.7	24.9	27.1	24.3	24.2	20.5	17.9
MCWBR		5.1	4.9	4.8	5.6	5.6	7.3	8.6	8.9	9.0	6.8	6.8	6.5	5.8
Clear-Sky Solar Irradiance	taub	0.409	0.404	0.391	0.372	0.352	0.331	0.330	0.336	0.393	0.420	0.436	0.427	
	taud	2.477	2.492	2.522	2.532	2.516	2.535	2.524	2.490	2.356	2.316	2.326	2.398	
	Ebn at Noon	297	296	293	287	282	283	286	294	287	287	287	292	
	Edh at Noon	37	36	34	32	30	29	30	33	40	43	43	41	
All-Sky Solar Radiation	RadAvg	1903	1917	1876	1839	1667	1547	1652	1909	2152	2250	2208	2036	
	RadStd	201	246	198	148	82	91	52	56	95	142	111	195	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page